

# Skills Summary

## Transformations with Multiple Parameters

Use this reference while completing your worksheet or when studying.

**The parameter map:**  $g(x) = a f(b(x - h)) + k$

$a$  **outside** vertical stretch by  $|a|$ ; reflect over the  $x$ -axis if  $a < 0$  does what it says

$k$  **outside** shift up  $k$  does what it says

$b$  **inside** horizontal stretch by  $1/b$  does the opposite

$h$  **inside** shift right  $h$  does the opposite

**Factor first.**  $h$  can only be read once the inside is written as  $b(x - h)$ .

### Skill 1 — The input chain

**Order of operations on a single input:**  $x \rightarrow (x - h) \rightarrow b(x - h) \rightarrow f(\dots) \rightarrow a \cdot f(\dots) \rightarrow a \cdot f(\dots) + k$ .

Build a column for each arrow. The parent function runs in the *middle* — everything to its left is an input move, everything to its right is an output move.

**Example:**  $f(x) = x^2$ ,  $g(x) = 2f(x - 1) + 3$ . Find  $g(4)$ .

**Step 1.** Two parameters act, one inside and one outside, so I walk the input through the chain in order rather than guessing at the end.

**Step 2.** Input move:  $4 - 1 = 3$ . Parent:  $f(3) = 9$ . Output moves:  $2(9) = 18$ , then  $18 + 3 = 21$ .  $\Rightarrow$   **$g(4) = 21$**

**Try it:**  $f(x) = x^2$ ,  $h(x) = -f(x + 2) + 6$ . Find  $h(0)$ .

**check:**  $h(0) = 6$

### Skill 2 — Factor before you read

**Rule:** factor  $b$  out of the *entire* inside before naming the shift —  $bx + c = b(x + \frac{c}{b})$ , so the shift is  $c/b$ , not  $c$ .

For  $y = A \sin(B(x - C)) + D$ : amplitude  $|A|$ , period  $2\pi/B$ , phase shift  $C$ , midline  $y = D$ .

**Example:** Rewrite  $y = 2 \sin(4x - \pi)$  in the form  $A \sin(B(x - C))$ .

**Step 1.** The shift is unreadable while 4 still multiplies only part of the inside, so the first move is to factor  $B = 4$  out of everything.

**Step 2.**  $4x - \pi = 4(x - \frac{\pi}{4})$ , since  $4 \cdot \frac{\pi}{4} = \pi$ . So  $A = 2$ ,  $B = 4$ ,  $C = \frac{\pi}{4}$ : shifted  $\frac{\pi}{4}$  right, not  $\pi$ .  $\Rightarrow$   **$2 \sin(4(x - \frac{\pi}{4}))$**

**Try it:** Rewrite  $y = 5 \sin(2x - \frac{\pi}{3})$  in the form  $A \sin(B(x - C))$ .

**check:**  $5 \sin(2(x - \frac{\pi}{6}))$

### Skill 3 — Composition order

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**Rule:** in  $(u \circ v)(x) = u(v(x))$  the *inner* map runs first. Swapping the order changes the function, because a stretch applied after a shift scales that shift, while a stretch applied before it does not.

**Example:**  $s(x) = x + 3$  (shift),  $d(x) = 4x$  (stretch). Simplify  $(d \circ s)(x)$ .

**Step 1.** The inner map  $s$  runs first, so the shift happens and the stretch then acts on the already-shifted value.

**Step 2.**  $d(s(x)) = d(x + 3) = 4(x + 3)$ . Distributing shows the shift was scaled: the graph moves 12, not 3.  $\Rightarrow$   **$4x + 12$**

**Try it:**  $t(x) = x + 2$ ,  $w(x) = 5x$ . Simplify  $(w \circ t)(x)$ .

**$5x + 10$**  ✓

**(!) Watch out:** Reading a shift straight off an unfactored inside is the most common multi-parameter error — always divide the constant by  $b$  (or factor, which does it for you). And when you undo parameters to solve an equation, peel them off from the *outside in*:  $k$  first, then  $a$ , then the parent, then  $b$  and  $h$ .